SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 06 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 40%

QUESTIONS: 15 Total Marks: 25

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

I M.Nivetha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Nivetha

Annexure A

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

(5 Marks)

Given:

- Box A: 6 Red, 4 Blue, 2 Green (12 balls)
- Box B: 5 Red, 7 Blue, 3 Green (15 balls)
- Box C: 8 Red, 2 Blue, 5 Green (15 balls)

Box selection probabilities:

- $P(A) = 1/4$
- $P(B) = 1/2$
- $P(C) = 1/4$

Blue ball probabilities:

- $P(Blue \mid A) = 4/12 = 1/3$
- $P(Blue \mid B) = 7/15$
- $P(Blue \mid C) = 2/15$

Using Bayes' Theorem:

$$P(C \mid Blue) = [P(C) \times P(Blue \mid C)] / P(Blue)$$

First find P(Blue):

$$P(Blue) = (1/4 \times 1/3) + (1/2 \times 7/15) + (1/4 \times 2/15)$$

$$= 1/12 + 7/30 + 1/30$$

$$= 7/20$$

Now,

=

$P(C \mid Blue) = (1/4 \times 2/15) \div (7/20)$

$= 1/30 \times 20/7$

$2/21$

Probability that the selected box was Box C = 2/21 ≈ 0.0952 (9.52%)

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C? (5 Marks)

Given:

- Section A: 5 Red, 3 Blue, 2 Black (10 cars)
- Section B: 4 Red, 6 Blue, 5 Black (15 cars)
- Section C: 7 Red, 2 Blue, 6 Black (15 cars)

Section selection probabilities:

- $P(A) = 1/4$
- $P(B) = 1/2$
- $P(C) = 1/4$

Blue car probabilities:

- $P(Blue \mid A) = 3/10$
- $P(Blue \mid B) = 6/15 = 2/5$
- $P(Blue \mid C) = 2/15$

Using Bayes' Theorem:

$P(C \mid Blue) = [P(C) \times P(Blue \mid C)] / P(Blue)$

First find P(Blue):

$P(Blue) = (1/4 \times 3/10) + (1/2 \times 2/5) + (1/4 \times 2/15)$

$= 3/40 + 1/5 + 1/30$

$= 9/120 + 24/120 + 4/120$

$= 37/120$

Now,

$P(C \mid \text{Blue}) = (1/4 \times 2/15) \div (37/120)$

=

1/30 × 120/37

= 4/37

Probability that the car came from Section C = 4/37 ≈ 0.1081 (10.81%)

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis. (5 Marks)Given:

- Total samples = 1800 + 1800 = 3600
- Test data = 20% of 3600 = 720 samples
- True Positives (TP) = 210
- True Negatives (TN) = 190
- False Positives (FP) = 30

TP + TN + FP + FN = 720

210 + 190 + 30 + FN = 720

430 + FN = 720

FN = 290

Accuracy = (TP + TN) / (TP + TN + FP + FN)

= (210 + 190) / 720

= 400 / 720

= 0.5556 = 55.56%

Precision = TP / (TP + FP)

= 210 / (210 + 30)

= 210 / 240

= 0.875 = 87.5%

Sensitivity = TP / (TP + FN)

= 210 / (210 + 290)

210 / 500

= 0.42 = 42%

Specificity = TN / (TN + FP)

= 190 / (190 + 30)

= 190 / 220

= 0.8636 = 86.36%

False Negatives (FN) = 290

Accuracy = 55.56%

Precision = 87.5%

Sensitivity (Recall) = 42%

Specificity = 86.36%

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

(a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
(b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
(d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
(h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i) Identify the epoch at which the model begins to overfit.

Overfitting begins at Epoch 4, because the training loss continues to decrease while the validation loss starts increasing.

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ii) Suggest two effective techniques to reduce overfitting in this model.

Two techniques to reduce overfitting:

- Apply Early Stopping.
- Use Dropout Regularization.

iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Effect of overfitting:

Overfitting causes the model to learn the training data too closely, including noise. As a result, it performs well on training data but poorly on unseen test data, leading to reduced generalization and lower prediction accuracy.

Overfitting starts at Epoch 4.
Techniques: Early Stopping and Dropout Regularization.
Overfitting reduces the model's ability to generalize, resulting in poor performance on new data.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)
(a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
(b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63% (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
(d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
(e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
(f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
(g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70% (h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68%

i) Identify the epoch at which the model starts overfitting.

The model starts overfitting at Epoch 6, because training accuracy continues to increase while validation accuracy starts decreasing.

ii) Explain why validation accuracy decreases despite training accuracy increasing.

Validation accuracy decreases because the model begins to memorize the training data instead of learning general patterns, resulting in poor performance on unseen data.

iii) Suggest two methods to improve validation performance.

Two methods to improve validation performance:

- Apply Early Stopping.
- Use Data Augmentation (or Dropout Regularization).

6. A machine learning model shows the following loss values during training:

(5 Marks)

(a) Epoch 1: Training Loss = 3.0, Validation Loss = 2.7
(b) Epoch 2: Training Loss = 2.6, Validation Loss = 2.3

(c) Epoch 3: Training Loss = 2.3, Validation Loss = 2.1
(d) Epoch 4: Training Loss = 2.1, Validation Loss = 2.0
(e) Epoch 5: Training Loss = 1.9, Validation Loss = 2.2
(f) Epoch 6: Training Loss = 1.7, Validation Loss = 2.5
(g) Epoch 7: Training Loss = 1.6, Validation Loss = 2.8
(h) Epoch 8: Training Loss = 1.4, Validation Loss = 3.1
i) At which epoch does the model achieve optimal generalization?

The model achieves optimal generalization at Epoch 4, where the validation loss is the lowest (2.0).

ii) Identify signs of overfitting from the data.

Overfitting starts after Epoch 4, as the training loss continues to decrease while the validation loss increases.

iii) Recommend two regularization techniques and explain their impact.

Two regularization techniques:

- Dropout Regularization – Randomly deactivates neurons during training, reducing overfitting.
- L2 Regularization (Weight Decay) – Penalizes large weights, improving generalization.

7. In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. Let the number of tires are 250 (125 front and 125 back tires).

(5 Marks)

a) Front identified as Front: 119
b) Front identifies as Back: 06
c) Back identified as Back: 120
d) Back identified as Front: 05

Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

Given:

- True Positive (TP) = 119
- False Negative (FN) = 6
- True Negative (TN) = 120
- False Positive (FP) = 5
- Total = 250

Accuracy

Accuracy = (TP + TN) / (TP + TN + FP + FN)

=

=

(119 + 120) / 250

239 / 250

= 0.956 = 95.6%

Precision

Precision = TP / (TP + FP)

= 119 / (119 + 5)

= 119 / 124

= 0.9597 = 95.97%

Sensitivity (Recall)

Sensitivity = TP / (TP + FN)

= 119 / (119 + 6)

= 119 / 125

= 0.952 = 95.2%

Specificity

Specificity = TN / (TN + FP)

= 120 / (120 + 5)

= 120 / 125

= 0.96 = 96.0%

F1-Score

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

= 2 × (0.9597 × 0.952) / (0.9597 + 0.952)

= 0.9558 = 95.58%

8. In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy): (5 Marks)

a) Diseased identified as Diseased: 138
b) Diseased identified as Healthy: 12
c) Healthy identified as Healthy: 135
d) Healthy identified as Diseased: 15

Compute: Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.

Given:

- True Positive (TP) = 138
- False Negative (FN) = 12
- True Negative (TN) = 135
- False Positive (FP) = 15
- Total = 300

Accuracy

Accuracy = (TP + TN) / (TP + TN + FP + FN)

= (138 + 135) / 300

= 273 / 300

= 0.91 = 91%

Precision

Precision = TP / (TP + FP)

= 138 / (138 + 15)

= 138 / 153

= 0.902 = 90.2% ensitivity

(Recall)

Sensitivity = TP / (TP + FN)

= 138 / (138 + 12)

= 138 / 150

= 0.92 = 92%

Specificity

Specificity = TN / (TN + FP)

= 135 / (135 + 15)

= 135 / 150

= 0.90 = 90%

F1-Score

=

=

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

2 × (0.902 × 0.92) / (0.902 + 0.92)

0.911 = 91.1%

9. In a spam email filtering system, an AI model classifies emails as Spam or Not Spam. Out of 400 emails (200 spam and 200 non-spam): (5 Marks)

- Spam identified as Spam: 180
- Spam identified as Not Spam: 20
- Not Spam identified as Not Spam: 170
- Not Spam identified as Spam: 30

Given:

- TP = 180
- FN = 20
- TN = 170
- FP = 30
- Total = 400 Accuracy

= (TP + TN) / Total

= (180 + 170) / 400

= 87.5%

Precision

= TP / (TP + FP)

= 180 / (180 + 30)

= 85.71%

Sensitivity (Recall)

= TP / (TP + FN)

= 180 / (180 + 20)

= 90%

Specificity

= TN / (TN + FP)

= 170 / (170 + 30)

= 85%

F1-Score

=

=

2 × (Precision × Recall) / (Precision + Recall)

= 87.8%

Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

10. In a quality inspection system in manufacturing, an AI model identifies defective and non-defective products. Out of 500 products (250 defective and 250 non-defective): (5 Marks)

- Defective identified as Defective: 2 30
- Defective identified as Non-Defective: 20
- Non-Defective identified as Non-Defective: 225
- Non-Defective identified as Defective: 25

Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

Given:

- TP = 230
- FN = 20
- TN = 225
- FP = 25
- Total = 500 Accuracy

= (TP + TN) / Total

= (230 + 225) / 500

= 91%

Precision

= TP / (TP + FP)

= 230 / (230 + 25)

= 230 / 255

= 90.20%

Sensitivity (Recall)

= TP / (TP + FN)

= 230 / (230 + 20)

230 / 250

=

92%

Specificity

= TN / (TN + FP)

= 225 / (225 + 25)

= 225 / 250

= 90%

F1-Score

= 2 × (Precision × Recall) / (Precision + Recall)

= 91.09%

11. In Shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in Shop Y, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with Shop Y being twice as likely to be chosen as Shop X. The selected bottle is found to be adulterated.

(5 Marks)

Given:

- P(X) = 1/3, P(Y) = 2/3
- P(Adulterated | X) = 30/50 = 3/5
- P(Adulterated | Y) = 20/60 = 1/3

P(Adulterated) = (1/3 × 3/5) + (2/3 × 1/3)

= 1/5 + 2/9

= 19/45

P(Y | Adulterated)

= (2/3 × 1/3) ÷ (19/45)

= 10/19 ≈ 52.63%

12. A warehouse has two sections. Section A contains 15 defective items and 35 non-defective items, while Section B contains 25 defective items and 25 non-defective items. One section is chosen at random, but Section A is three times as likely to be chosen as Section B. An item selected from the chosen section is found to be defective.

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(5 Marks)

Given:

- P(A) = 3/4, P(B) = 1/4
- P(Defective | A) = 15/50 = 3/10
- P(Defective | B) = 25/50 = 1/2

P(Defective)

= (3/4 × 3/10) + (1/4 × 1/2)

= 7/20

P(A | Defective)

= (3/4 × 3/10) ÷ (7/20)

= 9/14 ≈ 64.29%

13. In a binary classification problem using a deep learning model with a K-Nearest Neighbor classifier, the model produces True Positive (TP) = 120 and True Negative (TN) = 150.The dataset contains 1000 samples per class, and 20% of the data is used for testing.

(5 Marks)

Find the following: Accuracy, Precision, Sensitivity (Recall), Specificity.

Given:

- Total samples = 2000
- Test data = 20% = 400
- TP = 120, TN = 150
- FP + FN = 400 − (120 + 150) = 130

Assuming FP = FN = 65:

- Accuracy = 67.5%
- Precision = 64.86%
- Sensitivity = 64.86%
- Specificity = 69.77%

14. A medical image classification system using a deep learning framework combined with K-Nearest Neighbor (KNN) yields TP = 180 and TN = 160. Each class contains 1200 samples, and 25% of the dataset is used as test data.

Calculate: Accuracy Precision Sensitivity Specificity. (5 Marks)

Given:

- Total samples = 2400
- Test data = 25% = 600

- TP = 180, TN = 160
- FP + FN = 600 − (180 + 160) = 260

Assuming FP = FN = 130:

- Accuracy = 56.67%
- Precision = 58.06%
- Sensitivity = 58.06%
- Specificity = 55.17%

15. A deep learning model is trained for a classification task, and the following loss values are observed: (5 Marks)

i. Epoch 1: Training Loss = 2.8, Validation Loss = 2.6
ii. Epoch 2: Training Loss = 2.4, Validation Loss = 2.2
iii. Epoch 3: Training Loss = 2.1, Validation Loss = 1.9
iv. Epoch 4: Training Loss = 1.9, Validation Loss = 1.8
v. Epoch 5: Training Loss = 1.7, Validation Loss = 1.9
vi. Epoch 6: Training Loss = 1.5, Validation Loss = 2.1
vii. Epoch 7: Training Loss = 1.3, Validation Loss = 2.4
viii. Epoch 8: Training Loss = 1.2, Validation Loss = 2.7

Answer the following:

a. At which epoch does the model achieve the best generalization performance?

Best generalization performance: Epoch 4 (lowest validation loss = 1.8).

b. At which point does overfitting begin? Justify your answer.

Overfitting begins at Epoch 5, because the training loss keeps decreasing while the validation loss starts increasing.

c. Suggest two techniques to reduce overfitting and briefly explain how they help

Two techniques to reduce overfitting:

- Early Stopping – Stops training when validation loss starts increasing.
- Dropout Regularization – Randomly disables neurons during training to improve generalization.

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand